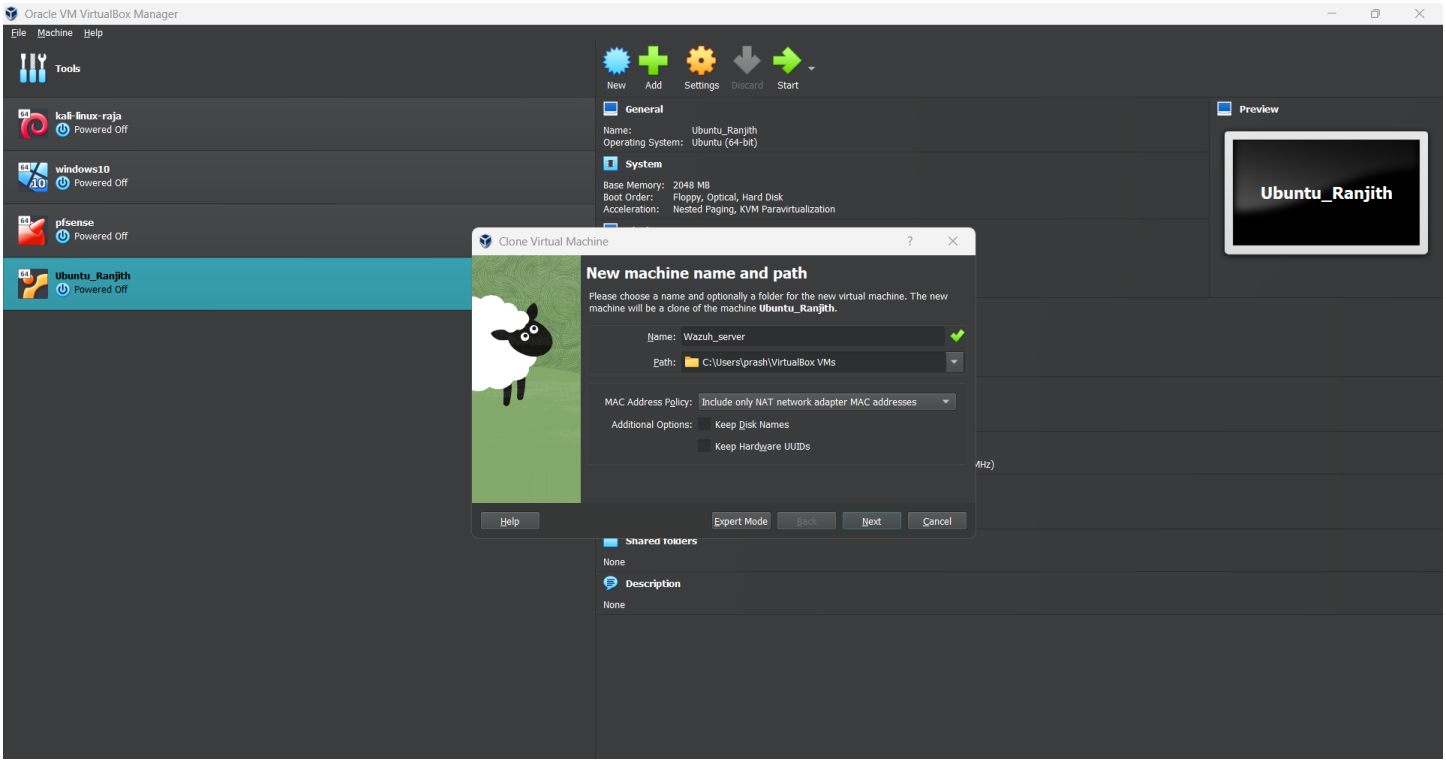
# CYBR125

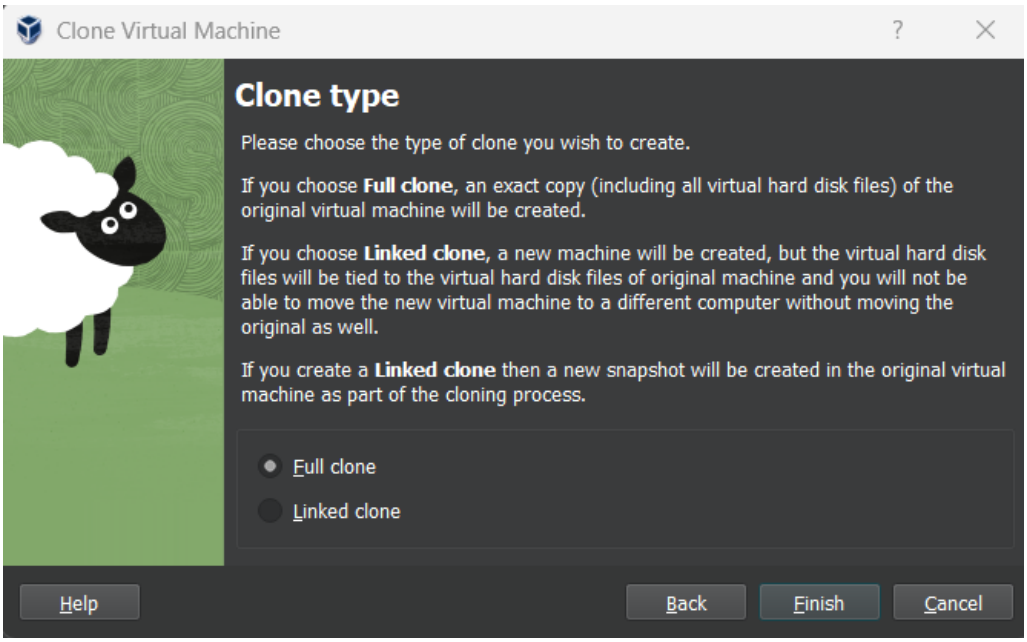
# Security Devices and Appliances

# LAB - 3

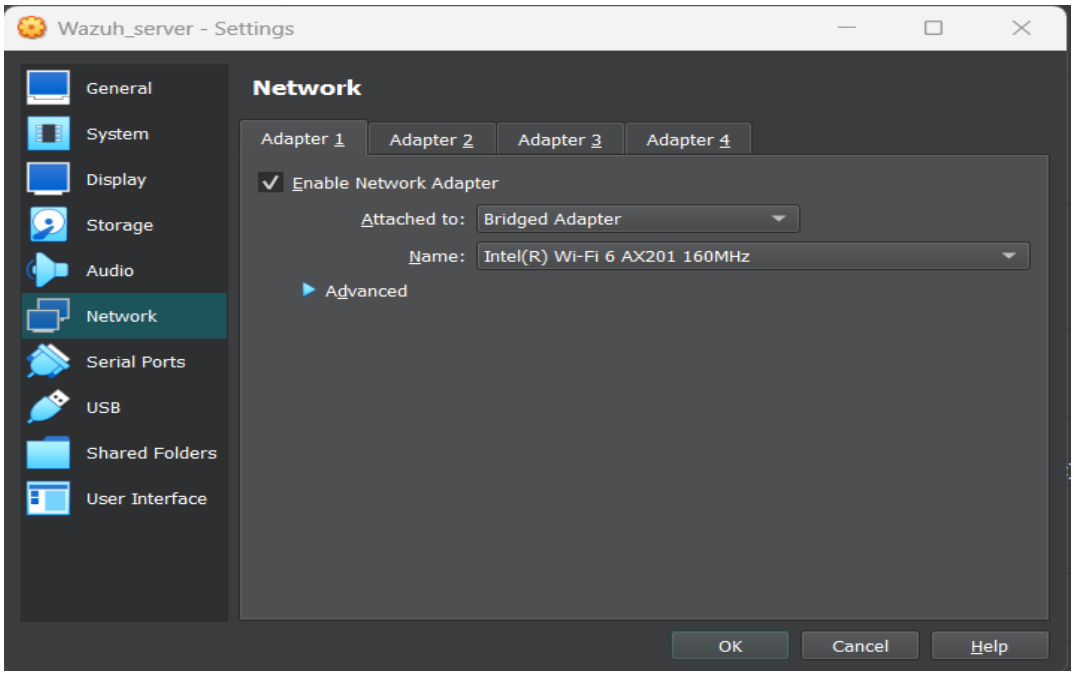
Ranjith kumar Murugan – A00191976

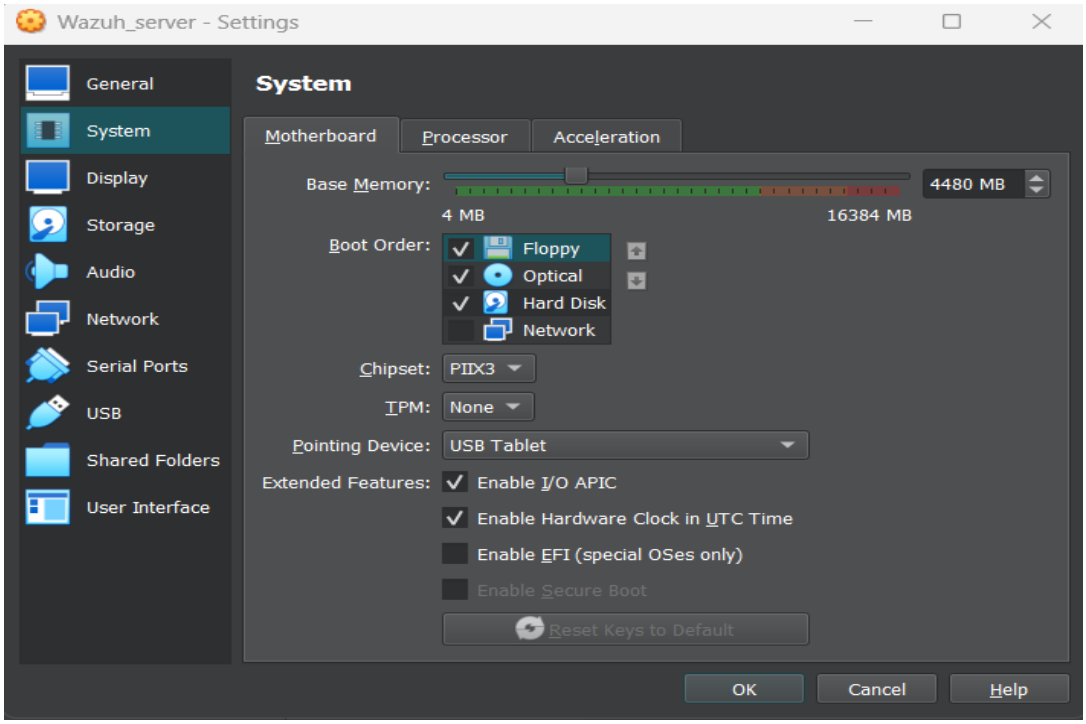
Download and Install the Ubuntu server. After the successful installation. Clone the Ubuntu server.

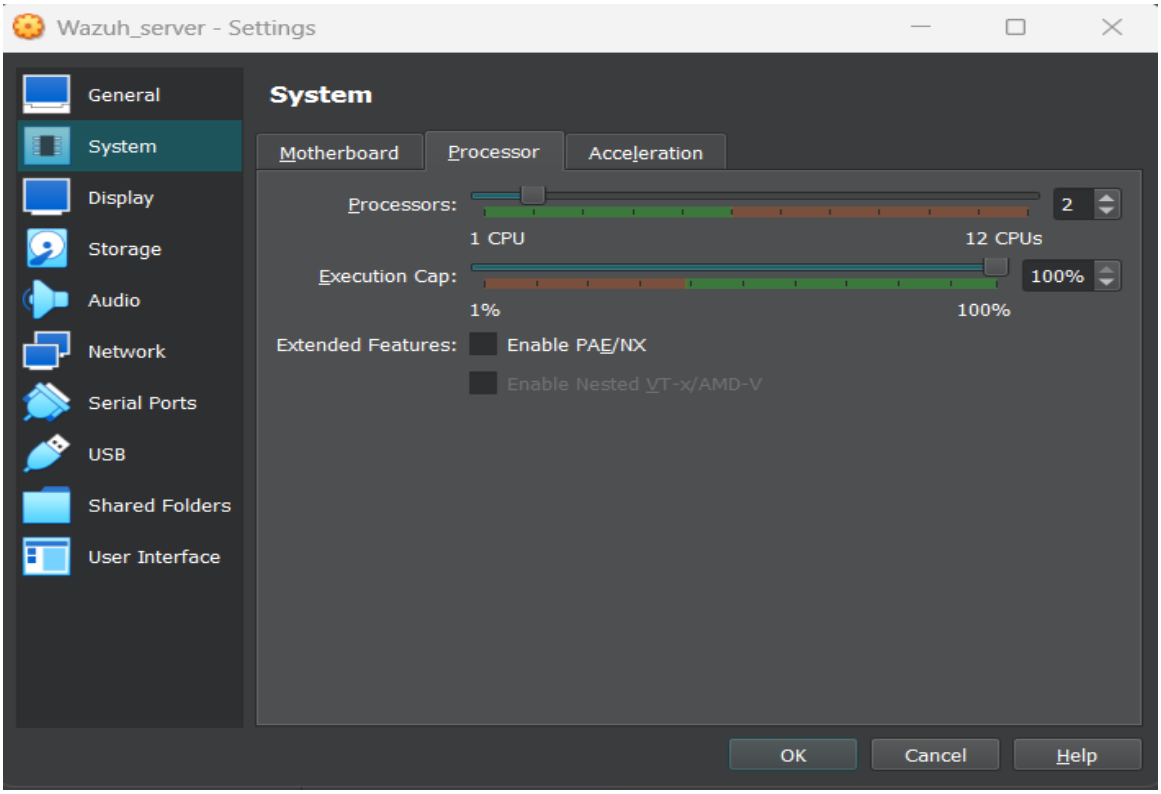


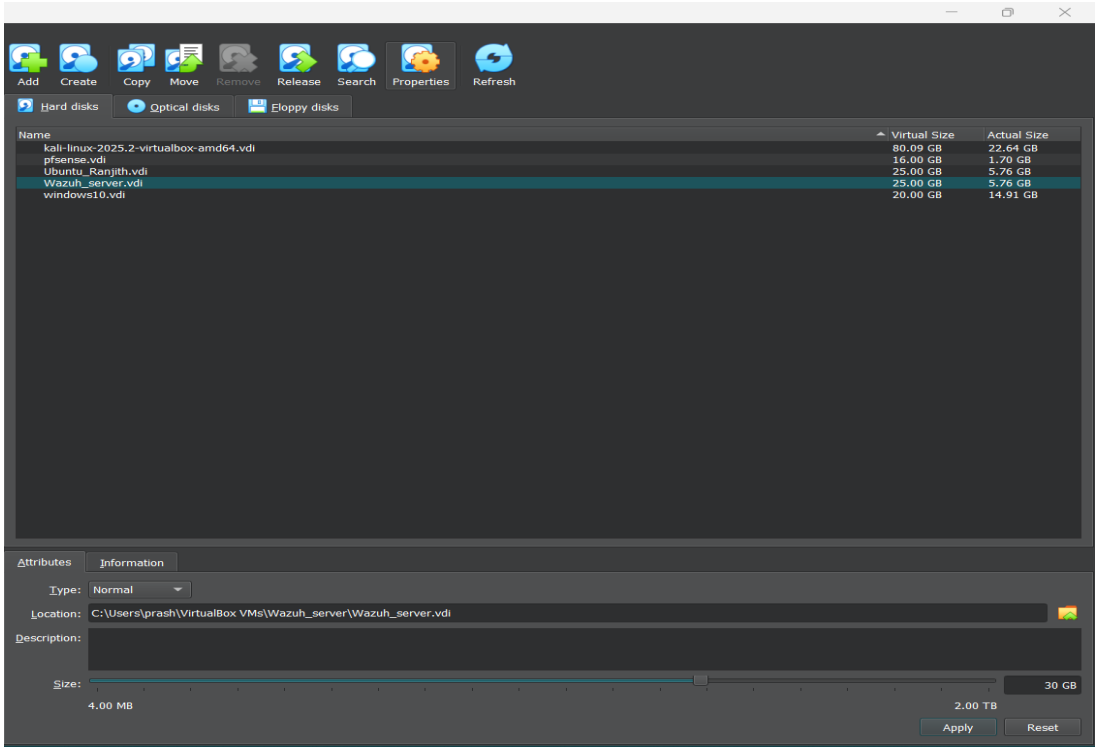



Set the server in Bridged network, and give 4GB Base Memory, 2 CPU with 30 GB storage. Click OK once you set.



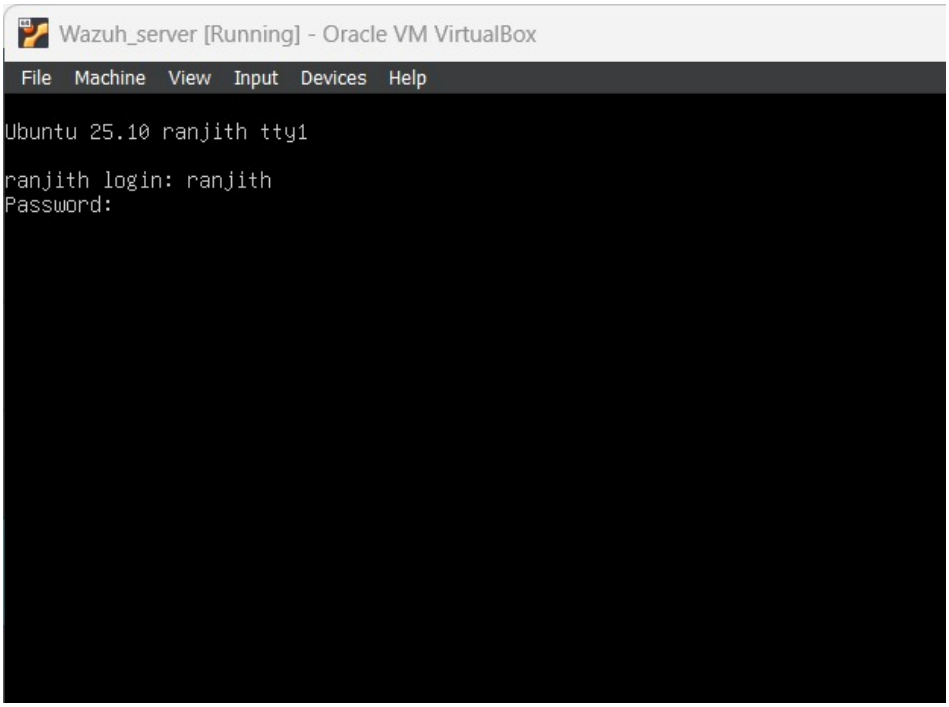




Wazuh_server - Settings
General
System
Display
Storage
Audio
Network
Serial Ports
USB
Shared Folders
User Interface
System
Motherboard
Processor
Acceleration
Processors:
2
1 CPU
12 CPUs
Execution Cap:
100%
1%
100%
Extended Features:
Enable PAE/NX
Enable Nested VT-x/AMD-V
OK
Cancel
Help

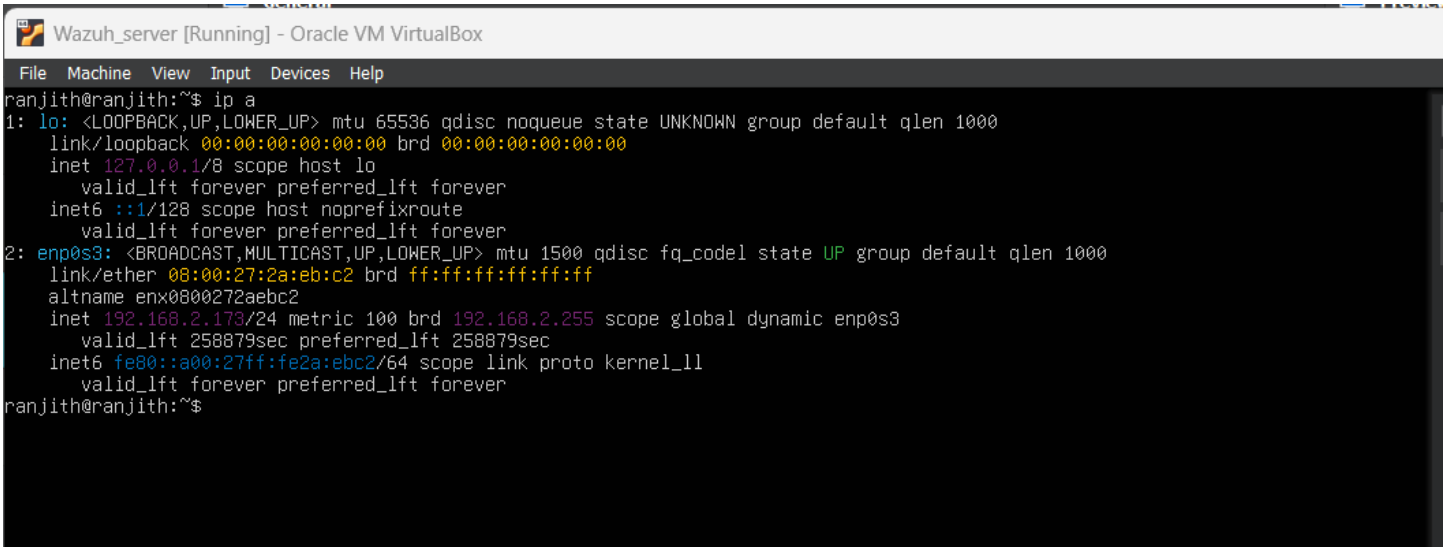

Add
Create
Copy
Move
Remove
Release
Search
Properties
Refresh
Hard disks
Optical disks
Floppy disks
Name
Virtual Size
Actual Size
kali-linux-2025.2-virtualbox-amd64.vdi 80.09 GB 22.64 GB
pfsense.vdi 16.00 GB 1.70 GB
Ubuntu_Ranjith.vdi 25.00 GB 5.76 GB
Wazuh_server.vdi 25.00 GB 5.76 GB
windows10.vdi 20.00 GB 14.91 GB
Attributes
Information
Type: Normal
Location: C:\Users\prash\VirtualBox VMs\Wazuh_server\Wazuh_server.vdi
Description:
Size:
30 GB
4.00 MB
2.00 TB
Apply
Reset

Start the server

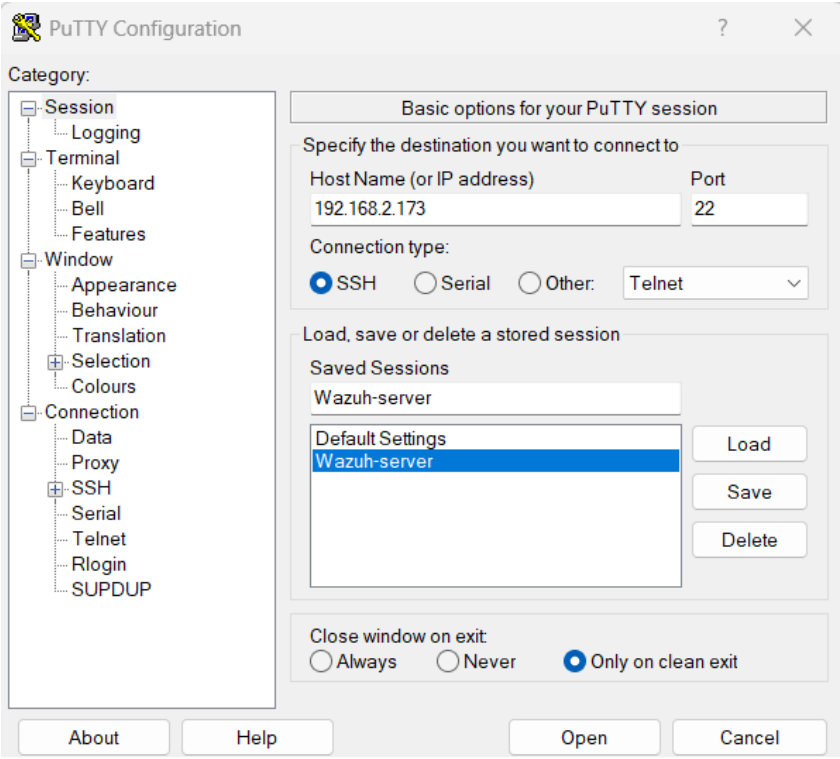
It will ask for login details

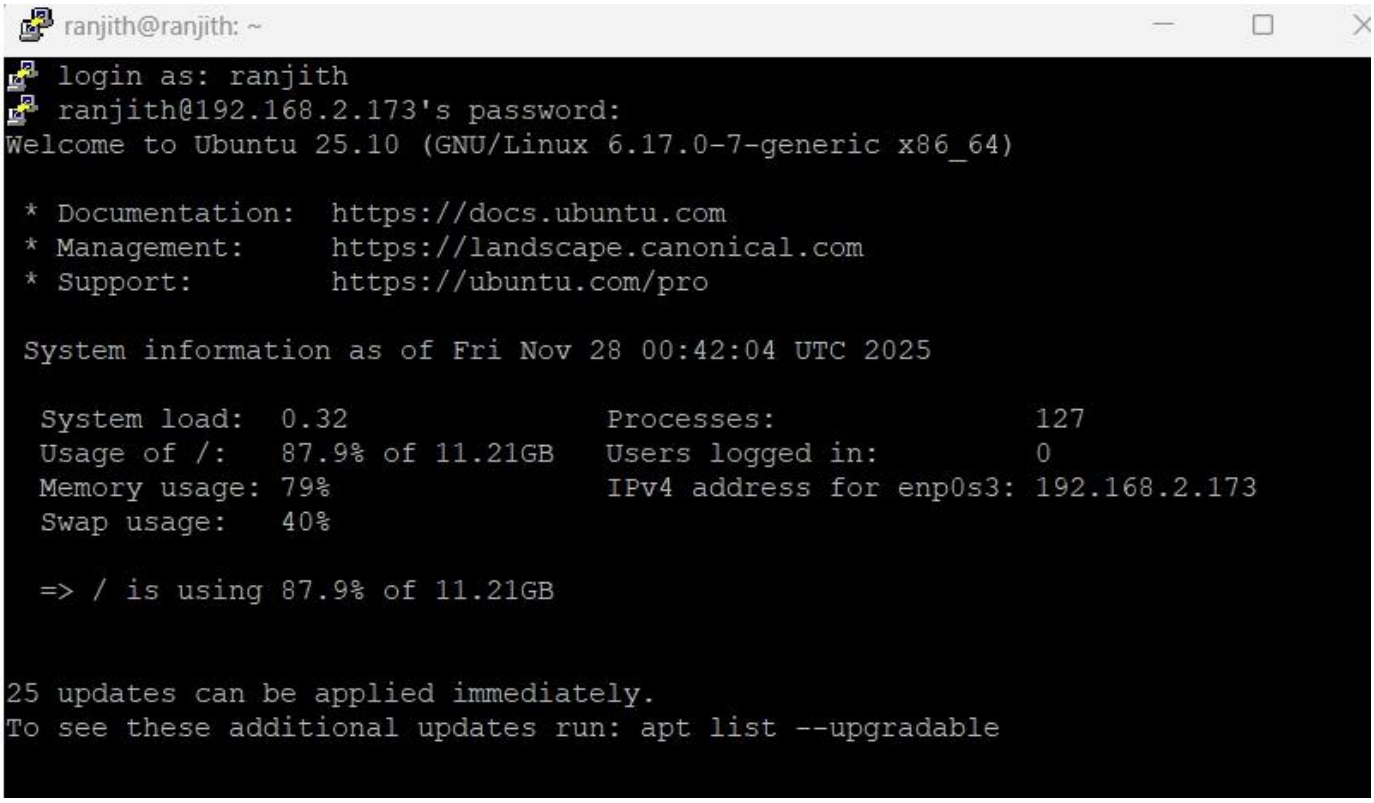



This is the Wazuh server IP address



Download PuTTY and Install. Open PuTTY connect the Wazuh server:



In server using sudo apt update && sudo apt upgrade -y

```
Get:4 http://ca.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:5 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,340 kB]
Ign:1 http://ca.archive.ubuntu.com/ubuntu noble InRelease
Get:6 http://ca.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,620 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.6 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [9,452 B]
Get:9 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:10 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [71.4 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [19.5 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
Get:13 http://ca.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]
Get:14 http://ca.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:15 http://ca.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1,500 kB]
Get:16 http://ca.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [378 kB]
Get:17 http://ca.archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [31.4 kB]
Get:18 http://ca.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:19 http://ca.archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7,128 B]
Get:20 http://ca.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [212 B]
Get:21 http://ca.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [11.0 kB]
Get:22 http://ca.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Hit:1 http://ca.archive.ubuntu.com/ubuntu noble InRelease
Fetched 5,565 kB in 3min 5s (30.0 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
33 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages will be upgraded:
  cloud-init coreutils distro-info-data fwupd gir1.2-packagekitglib-1.0 landscape-common libdrm-common libdrm2 libfwupd2
  libpam-systemd libsystemd-shared libsystemd0 libudev1 packagekit packagekit-tools powermgmt-base python3-software-prop
  software-properties-common sosreport systemd systemd-dev systemd-hwe-hwdb systemd-resolved systemd-sysv systemd-timesy
  ubuntu-pro-client ubuntu-pro-client-l10n udev
33 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
Need to get 51.8 MB of archives.
After this operation, 3,929 kB of additional disk space will be used.
Get:1 http://ca.archive.ubuntu.com/ubuntu noble-updates/main amd64 coreutils amd64 9.4-3ubuntu6.1 [1,413 kB]
Get:2 http://ca.archive.ubuntu.com/ubuntu noble-updates/main amd64 libnss-systemd amd64 255.4-1ubuntu8.11 [159 kB]
Get:3 http://ca.archive.ubuntu.com/ubuntu noble-updates/main amd64 systemd-dev all 255.4-1ubuntu8.11 [106 kB]
Get:4 http://ca.archive.ubuntu.com/ubuntu noble-updates/main amd64 systemd-timesyncd amd64 255.4-1ubuntu8.11 [35.3 kB]
Get:5 http://ca.archive.ubuntu.com/ubuntu noble-updates/main amd64 systemd-resolved amd64 255.4-1ubuntu8.11 [296 kB]
Get:6 http://ca.archive.ubuntu.com/ubuntu noble-updates/main amd64 libsystemd-shared amd64 255.4-1ubuntu8.11 [2,076 kB]
Get:7 http://ca.archive.ubuntu.com/ubuntu noble-updates/main amd64 libsystemd0 amd64 255.4-1ubuntu8.11 [434 kB]
Get:8 http://ca.archive.ubuntu.com/ubuntu noble-updates/main amd64 systemd-sysv amd64 255.4-1ubuntu8.11 [11.9 kB]
Get:9 http://ca.archive.ubuntu.com/ubuntu noble-updates/main amd64 libpam-systemd amd64 255.4-1ubuntu8.11 [235 kB]
Get:10 http://ca.archive.ubuntu.com/ubuntu noble-updates/main amd64 systemd amd64 255.4-1ubuntu8.11 [3,475 kB]
Get:11 http://ca.archive.ubuntu.com/ubuntu noble-updates/main amd64 udev amd64 255.4-1ubuntu8.11 [1,874 kB]
19% [11 udev 17.4 kB/1,874 kB 1%]
```

```
ranjith@wazuh:~$ sudo parted
[sudo] password for ranjith:
GNU Parted 3.6
Using /dev/sda
Welcome to GNU Parted! Type 'help' to view a list of commands.
(parted) print'
Warning: Not all of the space available to /dev/sda appears to be used, you can
fix the GPT to use all of the space (an extra 10485760 blocks) or continue with
the current setting?
Fix/Ignore? Fix
Model: ATA VBOX HARDDISK (scsi)
Disk /dev/sda: 32.2GB
Sector size (logical/physical): 512B/512B
Partition Table: gpt
Disk Flags:

Number  Start   End     Size    File system  Name  Flags
 1      1049kB  2097kB  1049kB                     bios_grub
 2      2097kB  2150MB  2147MB  ext4
 3      2150MB  26.8GB  24.7GB

(parted) resizepart
Partition number? 3
End?  [26.8GB]? 29GB
(parted) print
Model: ATA VBOX HARDDISK (scsi)
Disk /dev/sda: 32.2GB
Sector size (logical/physical): 512B/512B
Partition Table: gpt
Disk Flags:

Number  Start   End     Size    File system  Name  Flags
 1      1049kB  2097kB  1049kB                     bios_grub
 2      2097kB  2150MB  2147MB  ext4
```

```
(parted) print
Model: ATA VBOX HARDDISK (scsi)
Disk /dev/sda: 32.2GB
Sector size (logical/physical): 512B/512B
Partition Table: gpt
Disk Flags:

Number  Start   End     Size    File system  Name  Flags
 1      1049kB  2097kB  1049kB                     bios_grub
 2      2097kB  2150MB  2147MB  ext4
 3      2150MB  29.0GB  26.9GB

(parted) quit
```

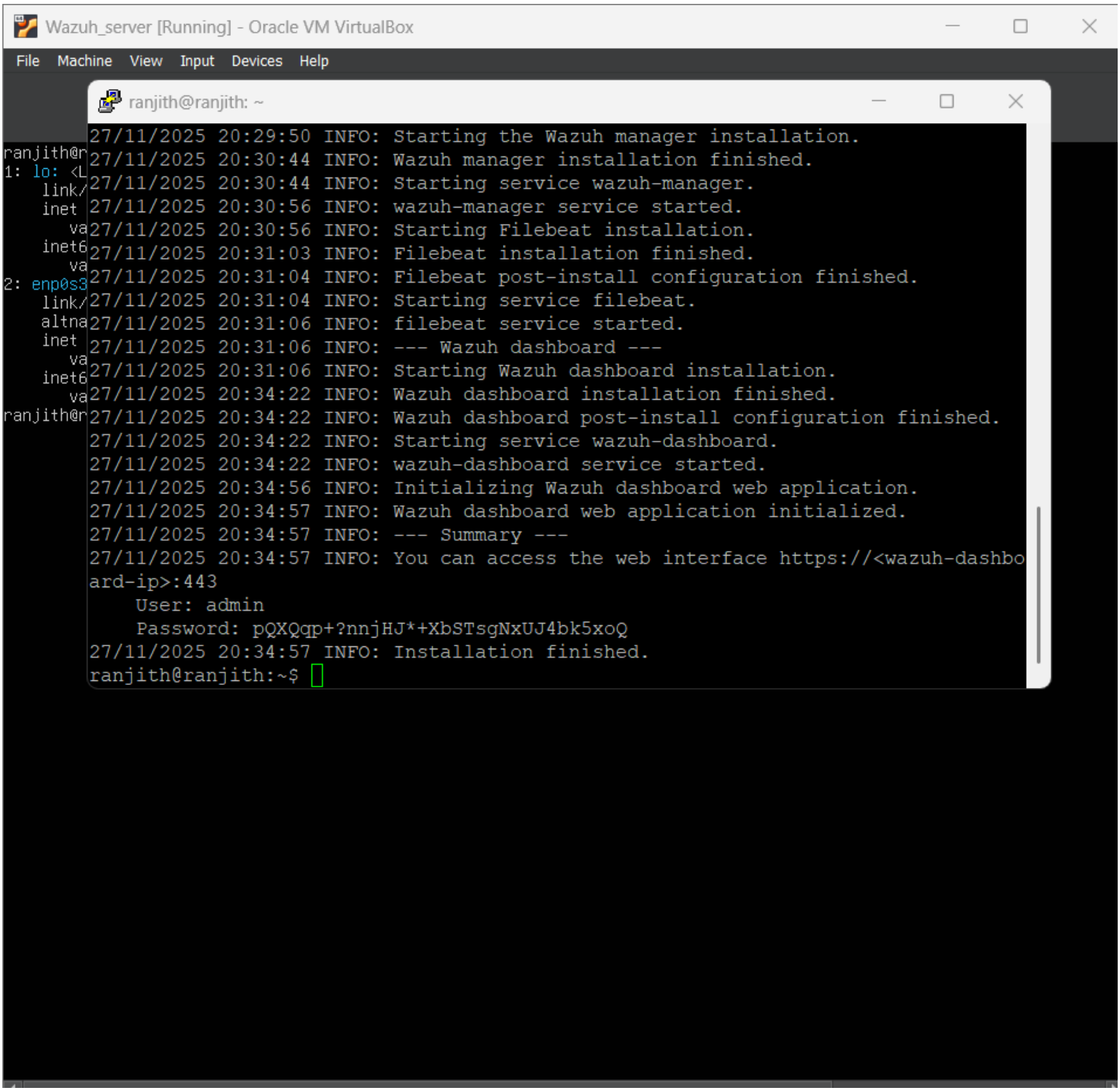
```
ranjith@wazuh:~$ sudo pvresize /dev/sda3
  Physical volume "/dev/sda3" changed
  1 physical volume(s) resized or updated / 0 physical volume(s) not resized
ranjith@wazuh:~$ sudo lvresize -l +100%FREE /dev/mapper/ubuntu--vg-ubuntu--lv
  Size of logical volume ubuntu-vg/ubuntu-lv changed from <11.50 GiB (2943 extents) to 25.00
GiB (6401 extents).
  Logical volume ubuntu-vg/ubuntu-lv successfully resized.
ranjith@wazuh:~$ sudo resize2fs /dev/mapper/ubuntu--vg-ubuntu--lv
resize2fs 1.47.0 (5-Feb-2023)
Filesystem at /dev/mapper/ubuntu--vg-ubuntu--lv is mounted on /; on-line resizing required
old_desc_blocks = 2, new_desc_blocks = 4
The filesystem on /dev/mapper/ubuntu--vg-ubuntu--lv is now 6554624 (4k) blocks long.

ranjith@wazuh:~$ df -h
Filesystem                         Size  Used Avail Use% Mounted on
tmpfs                              446M  1.1M  445M   1% /run
/dev/mapper/ubuntu--vg-ubuntu--lv   25G  4.6G   19G  20% /
tmpfs                              2.2G     0  2.2G   0% /dev/shm
tmpfs                              5.0M     0  5.0M   0% /run/lock
/dev/sda2                          2.0G  102M  1.7G   6% /boot
tmpfs                              446M   12K  446M   1% /run/user/1000
ranjith@wazuh:~$ lsblk
NAME                      MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda                         8:0    0   30G  0 disk
├─sda1                      8:1    0    1M  0 part
├─sda2                      8:2    0    2G  0 part /boot
└─sda3                      8:3    0   25G  0 part
  └─ubuntu--vg-ubuntu--lv 252:0    0   25G  0 lvm  /
sr0                        11:0    1 1024M  0 rom
ranjith@wazuh:~$ sudo reboot

Broadcast message from root@wazuh on pts/1 (Wed 2025-11-26 02:10:31 UTC):

The system will reboot now!
```

Again logged in the server:

```
uazuh login: ranjith
Password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-88-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Wed Nov 26 02:10:45 AM UTC 2025

  System load:  0.34               Processes:               119
  Usage of /:   18.5% of 24.50GB   Users logged in:         0
  Memory usage: 5%                 IPv4 address for enp0s3: 192.168.2.162
  Swap usage:   0%


Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
```

Install the Wazuh on the server using PuttY:

```
ranjith@ranjith: ~
ranjith@ranjith:~$ curl -sO https://packages.wazuh.com/4.7/wazuh-install.sh && s
udo bash ./wazuh-install.sh -a -i
27/11/2025 20:28:41 INFO: Starting Wazuh installation assistant. Wazuh version:
4.7.5
27/11/2025 20:28:41 INFO: Verbose logging redirected to /var/log/wazuh-install.l
og
27/11/2025 20:28:44 WARNING: Hardware and system checks ignored.
27/11/2025 20:28:44 INFO: Wazuh web interface port will be 443.
27/11/2025 20:28:46 INFO: --- Dependencies ----
27/11/2025 20:28:46 INFO: Installing apt-transport-https.
27/11/2025 20:28:49 INFO: Wazuh repository added.
27/11/2025 20:28:49 INFO: --- Configuration files ---
27/11/2025 20:28:49 INFO: Generating configuration files.
27/11/2025 20:28:49 INFO: Created wazuh-install-files.tar. It contains the Wazuh
 cluster key, certificates, and passwords necessary for installation.
27/11/2025 20:28:50 INFO: --- Wazuh indexer ---
27/11/2025 20:28:50 INFO: Starting Wazuh indexer installation.
```



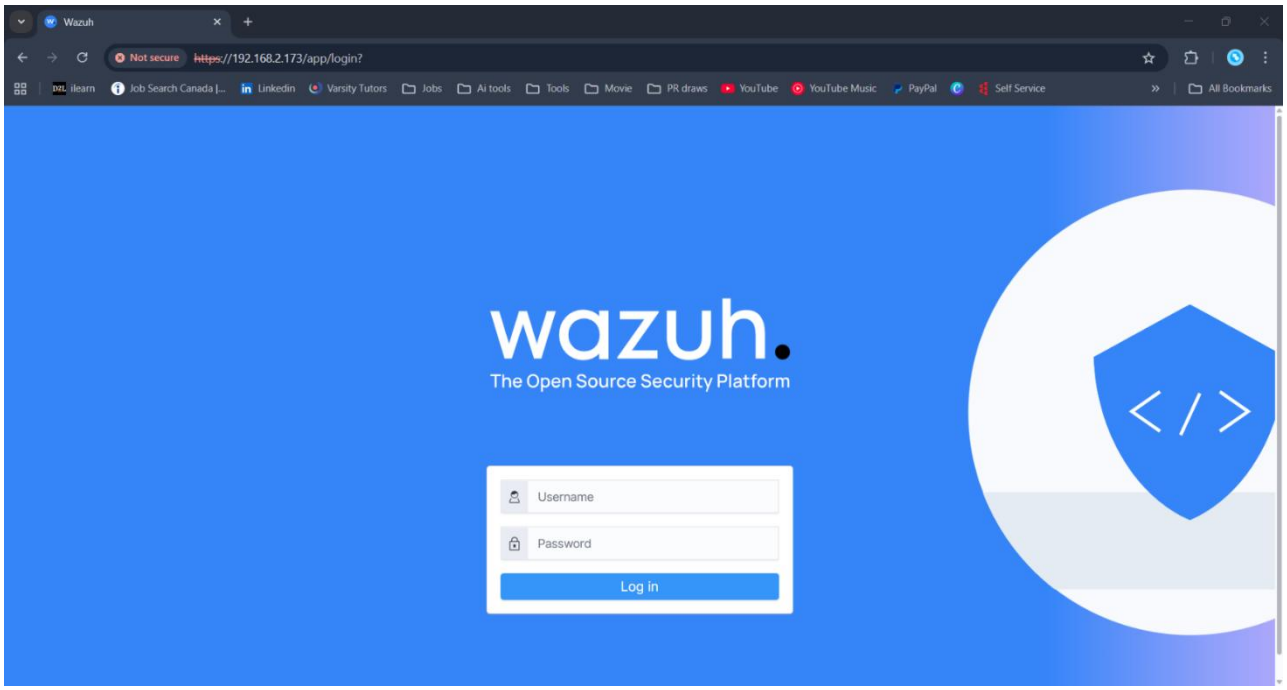

After successful installation it will give User and Password. Note that for to access the Wazuh Dashboard.

My Wazuh IP is: 192.168.2.173.

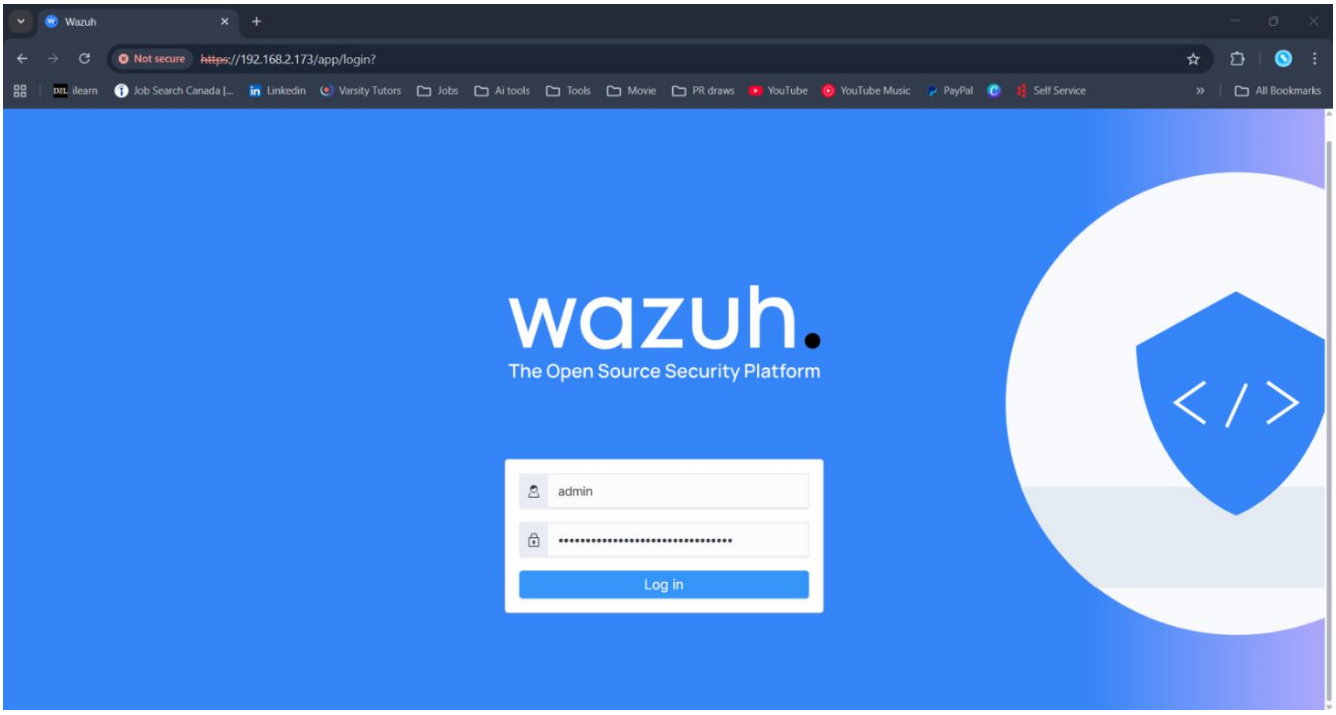
Go to Chrome and  go to https://<Wazuh server IP>

Username: admin

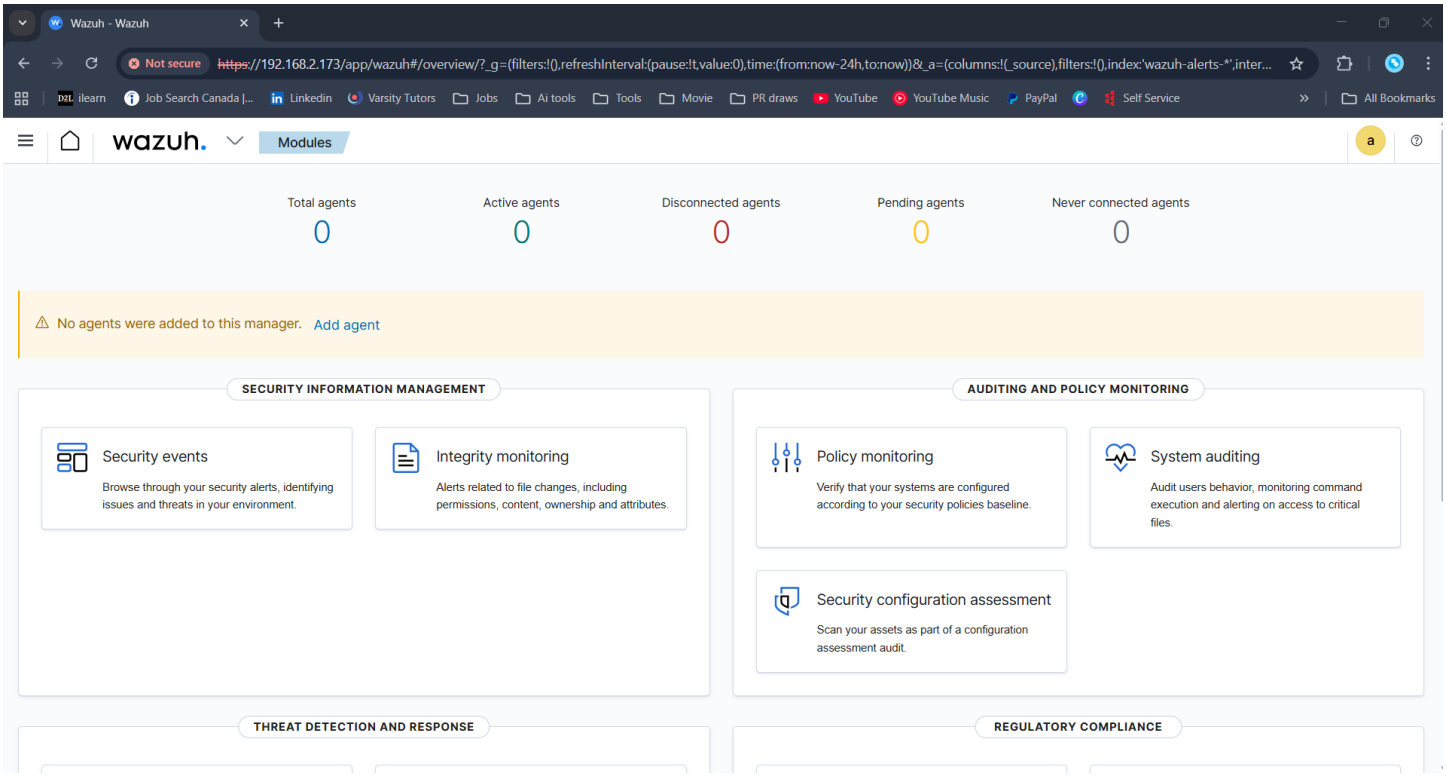
Password: your given password




Use the credentials

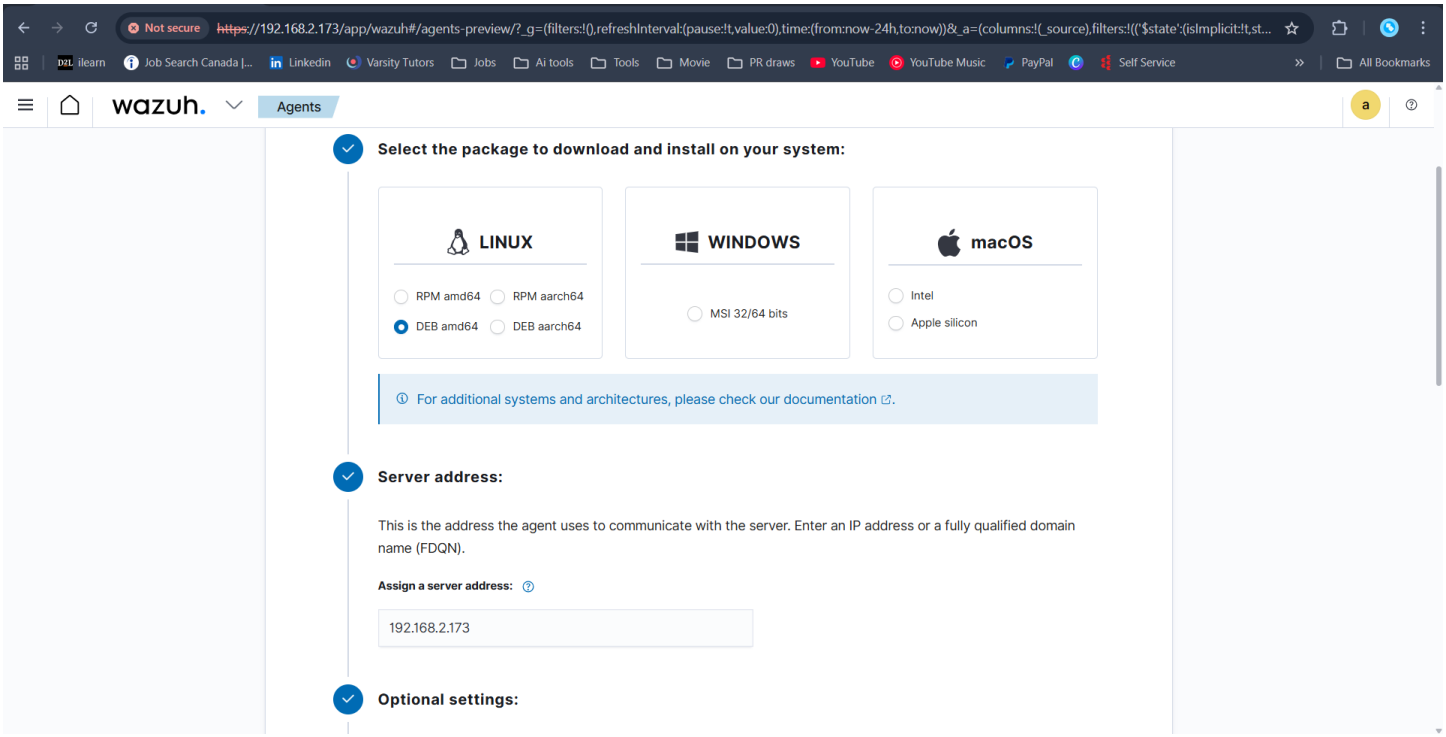


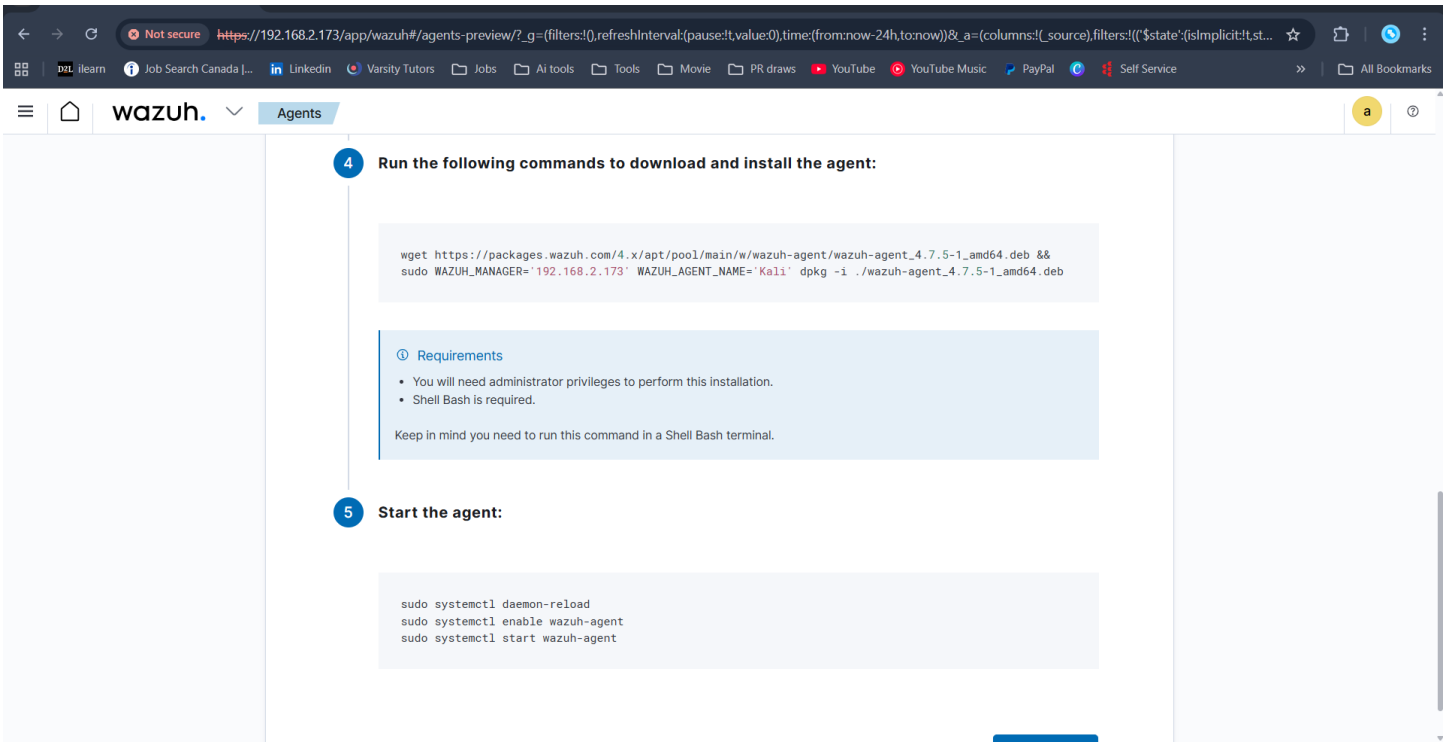
Post login of Wazuh:



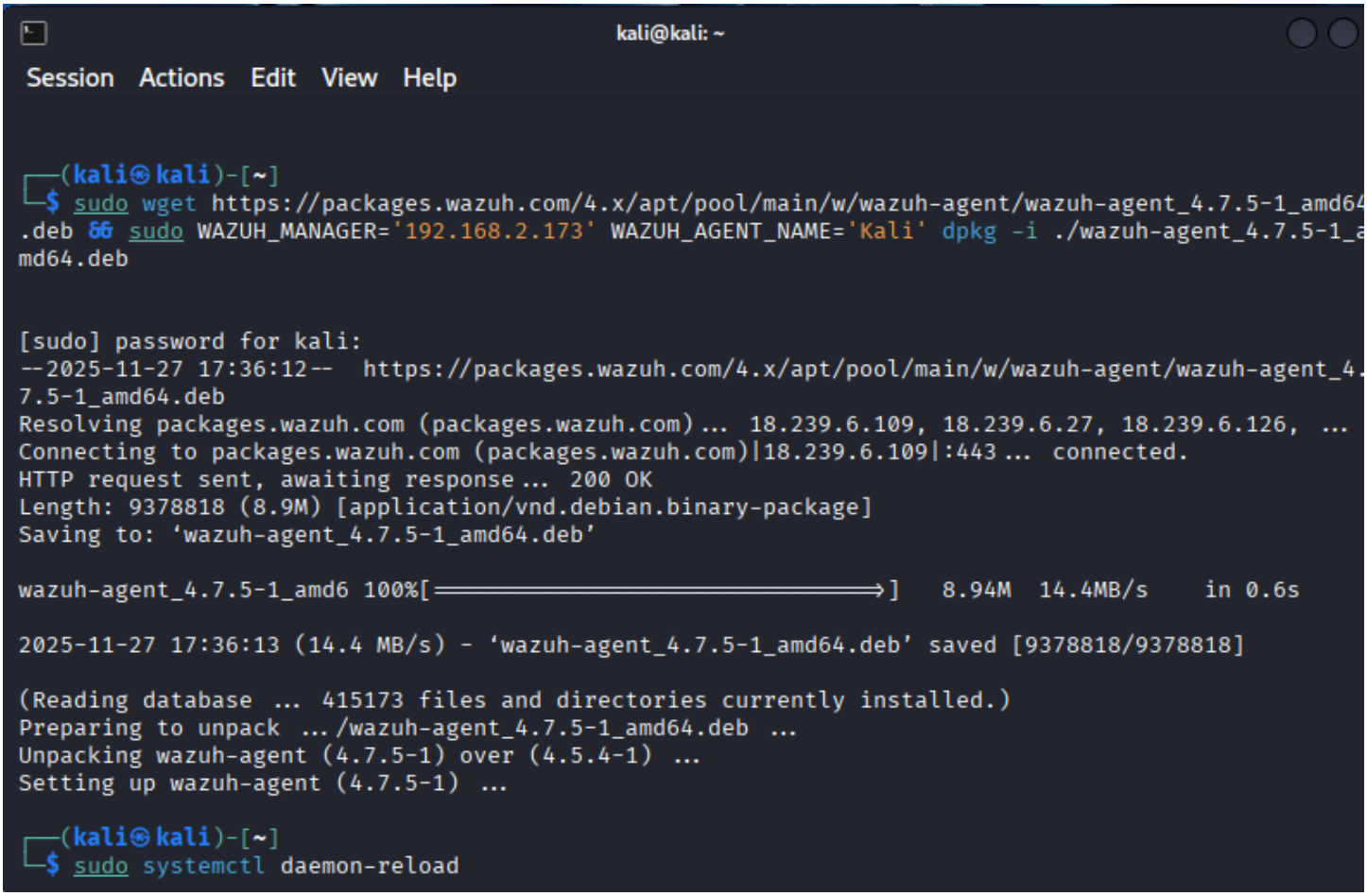

To add Agents Go to Add agent Option. Chose the OS version. Give the details.

Now I'm going to add the Kali OS to Wazuh.



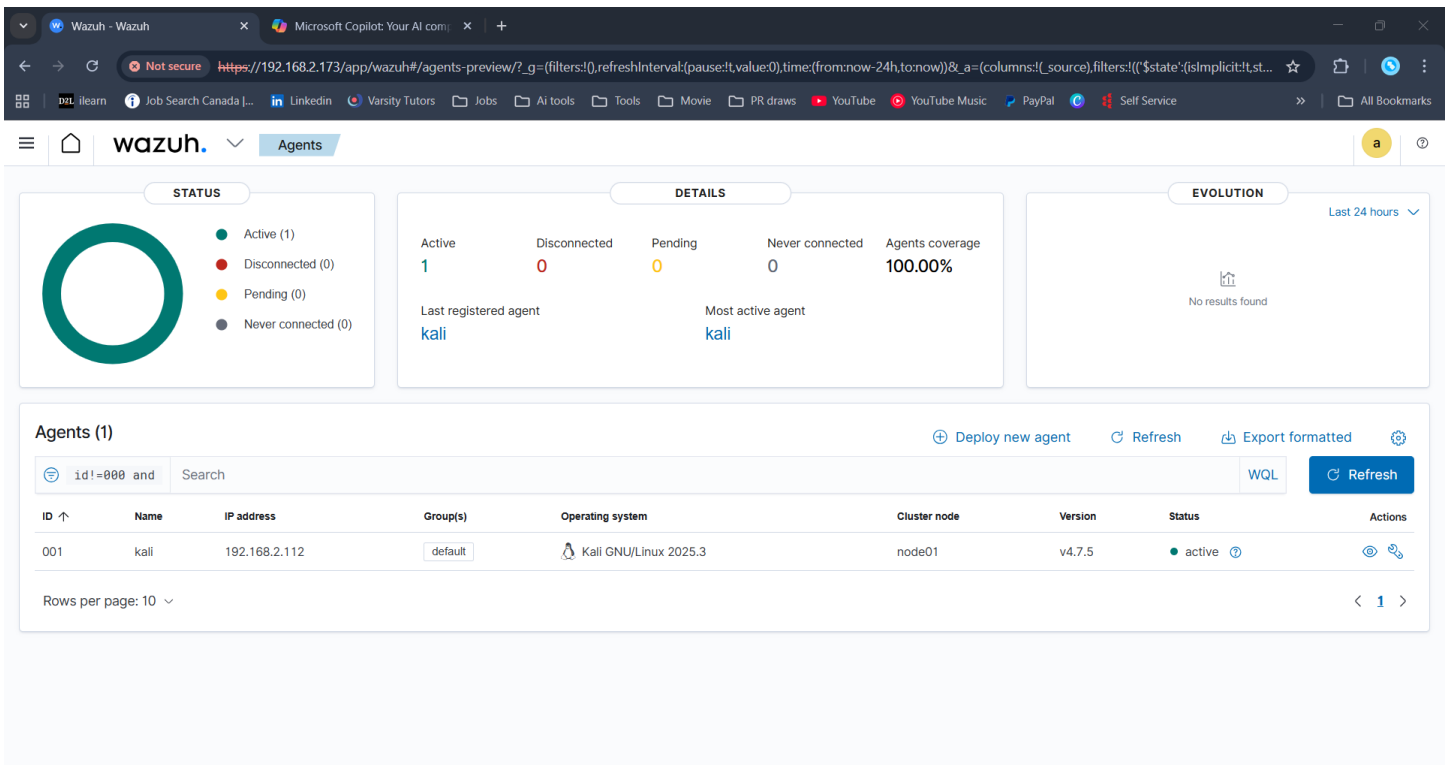



Run those commands in Kali terminal



```
Setting up wazuh-agent (4.7.5-1) ...

┌──(kali㉿kali)-[~]
└─$ sudo systemctl daemon-reload

┌──(kali㉿kali)-[~]
└─$ sudo systemctl enable wazuh-agent

┌──(kali㉿kali)-[~]
└─$ sudo systemctl start wazuh-agent

┌──(kali㉿kali)-[~]
└─$
```

Now We added Kali OS in the Wazuh SIEM.

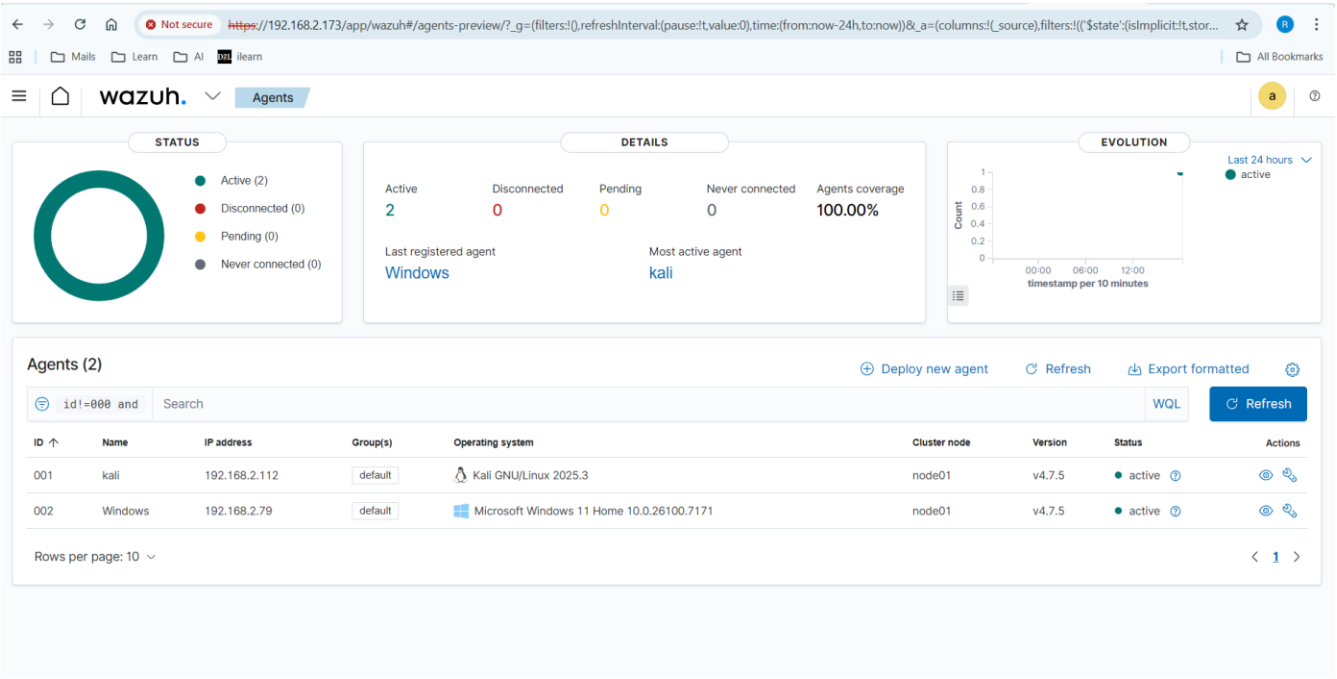


Now I'm adding the Windows agent to Wazuh.

Using Powershell add the Windows agent to Wazuh.

```
Select Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

                Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindow
s
                                                PS C:\WINDOWS\system32> Invoke-WebRequest -Uri https://p
ackages.wazuh.com/4.x/windows/wazuh-agent-4.7.5-1.msi -OutFile ${env.tmp}\wazuh-agent; msiexec.exe /i ${
env.tmp}\wazuh-agent /q WAZUH_MANAGER='192.168.2.173' WAZUH_AGENT_NAME='Windows' WAZUH_REGISTRATION_SERV
ER='192.168.2.173'
Invoke-WebRequest : Access to the path 'C:\wazuh-agent' is denied.
At line:1 char:1
+ Invoke-WebRequest -Uri https://packages.wazuh.com/4.x/windows/wazuh-a ...
+ ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
    + CategoryInfo          : NotSpecified: (:) [Invoke-WebRequest], UnauthorizedAccessException
    + FullyQualifiedErrorId : System.UnauthorizedAccessException,Microsoft.PowerShell.Commands.InvokeWeb
RequestCommand

PS C:\WINDOWS\system32> Invoke-WebRequest -Uri "https://packages.wazuh.com/4.x/windows/wazuh-agent-4.7.5
-1.msi" -OutFile "$env:TEMP\wazuh-agent.msi"
PS C:\WINDOWS\system32> msiexec.exe /i "$env:TEMP\wazuh-agent.msi" /q WAZUH_MANAGER="192.168.2.173" WAZU
H_AGENT_NAME="Windows" WAZUH_AGENT_GROUP="default"
PS C:\WINDOWS\system32> net start WazuhSvc

The Wazuh service was started successfully.

PS C:\WINDOWS\system32>
```

Now we successfully added windows agent.



**Vulnerability Scan :**

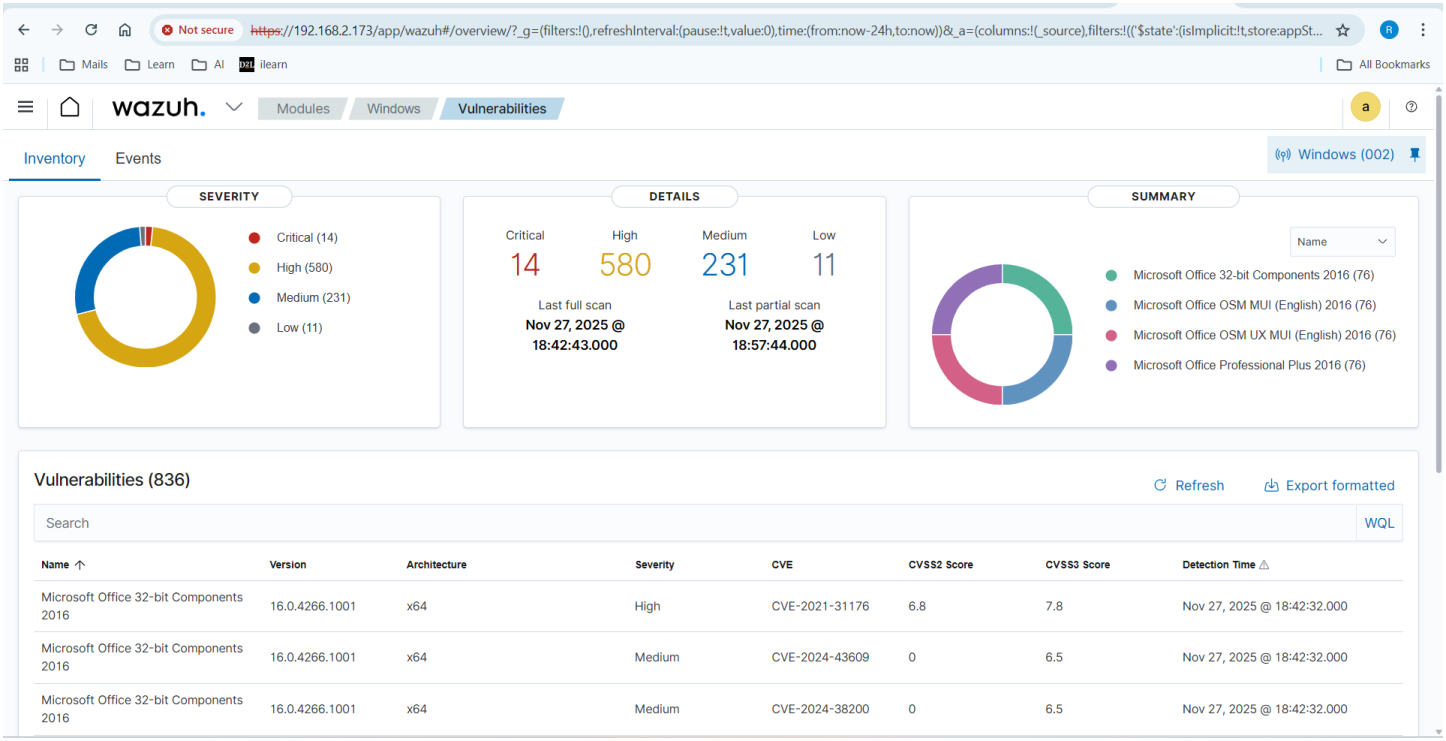
First, we need to enable the module

sudo nano /var/ossec/etc/ossec.conf

```
ossec_config>
 <global>
   <jsonout_output>yes</jsonout_output>
   <alerts_log>yes</alerts_log>
   <logall>no</logall>
   <logall_json>no</logall_json>
   <email_notification>no</email_notification>
   <smtp_server>smtp.example.wazuh.com</smtp_server>
   <email_from>wazuh@example.wazuh.com</email_from>
   <email_to>recipient@example.wazuh.com</email_to>
   <email_maxperhour>12</email_maxperhour>
   <email_log_source>alerts.log</email_log_source>
   <agents_disconnection_time>10m</agents_disconnection_time>
   <agents_disconnection_alert_time>0</agents_disconnection_alert_time>
 </global>

 <alerts>
   <log_alert_level>3</log_alert_level>
   <email_alert_level>12</email_alert_level>
 </alerts>

 <!-- Choose between "plain", "json", or "plain,json" for the format of internal logs -->
 <logging>
   <log_format>plain</log_format>
 </logging>

 <remote>
   <connection>secure</connection>
   <port>1514</port>
   <protocol>tcp</protocol>
   <queue_size>131072</queue_size>
 </remote>

 <!-- Policy monitoring -->
 <rootcheck>
   <disabled>no</disabled>
   <check_files>yes</check_files>
   <check_trojans>yes</check_trojans>
                                   [ Read 401 lines ]
G Help      ^O Write Out   ^F Where Is    ^K Cut       ^T Execute    ^C Location
X Exit      ^R Read File   ^\ Replace     ^U Paste     ^J Justify    ^/ Go To Line
```

```
<vulnerability-detector>
 <enabled>yes</enabled>
 <interval>5m</interval>
 <min_full_scan_interval>6h</min_full_scan_interval>
 <run_on_start>yes</run_on_start>
</vulnerability-detector>
```

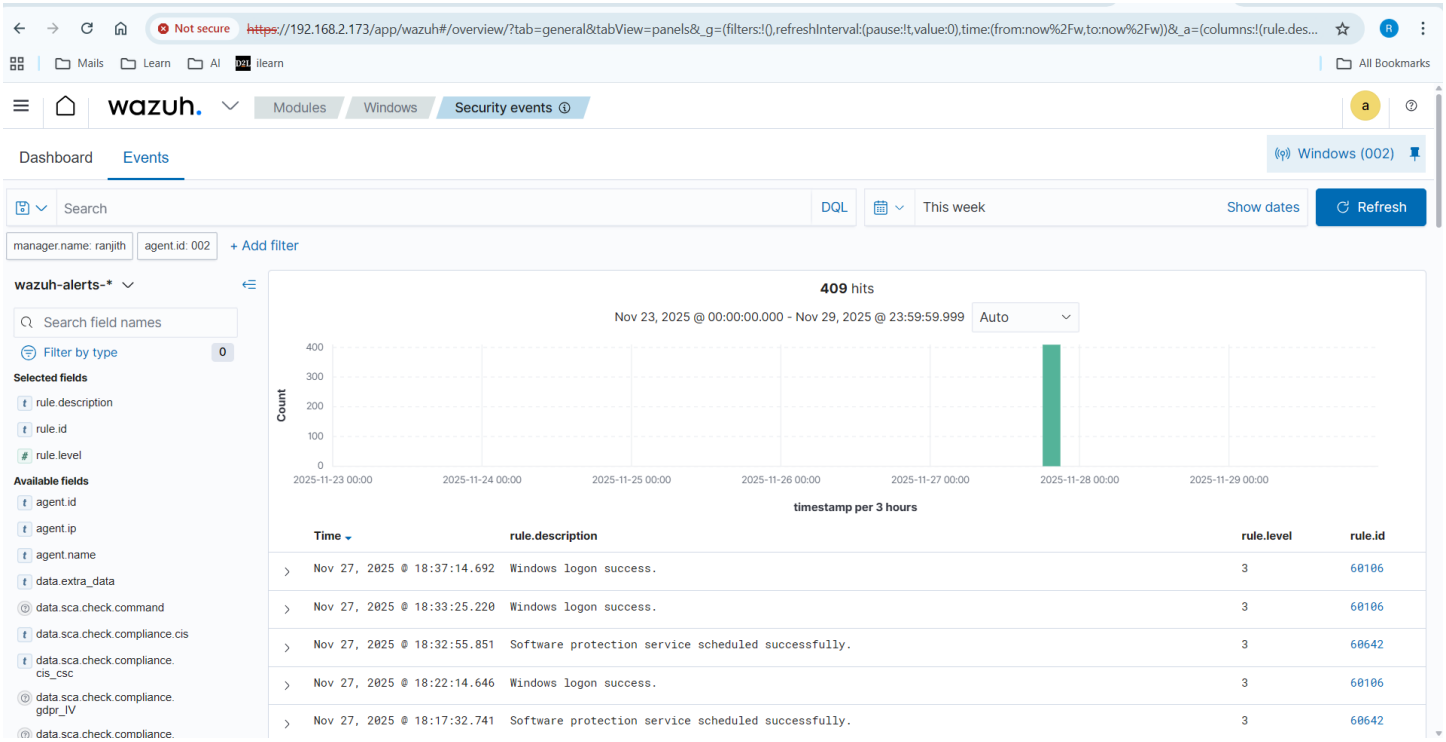
Wazuh → Agents → Vulnerabilities

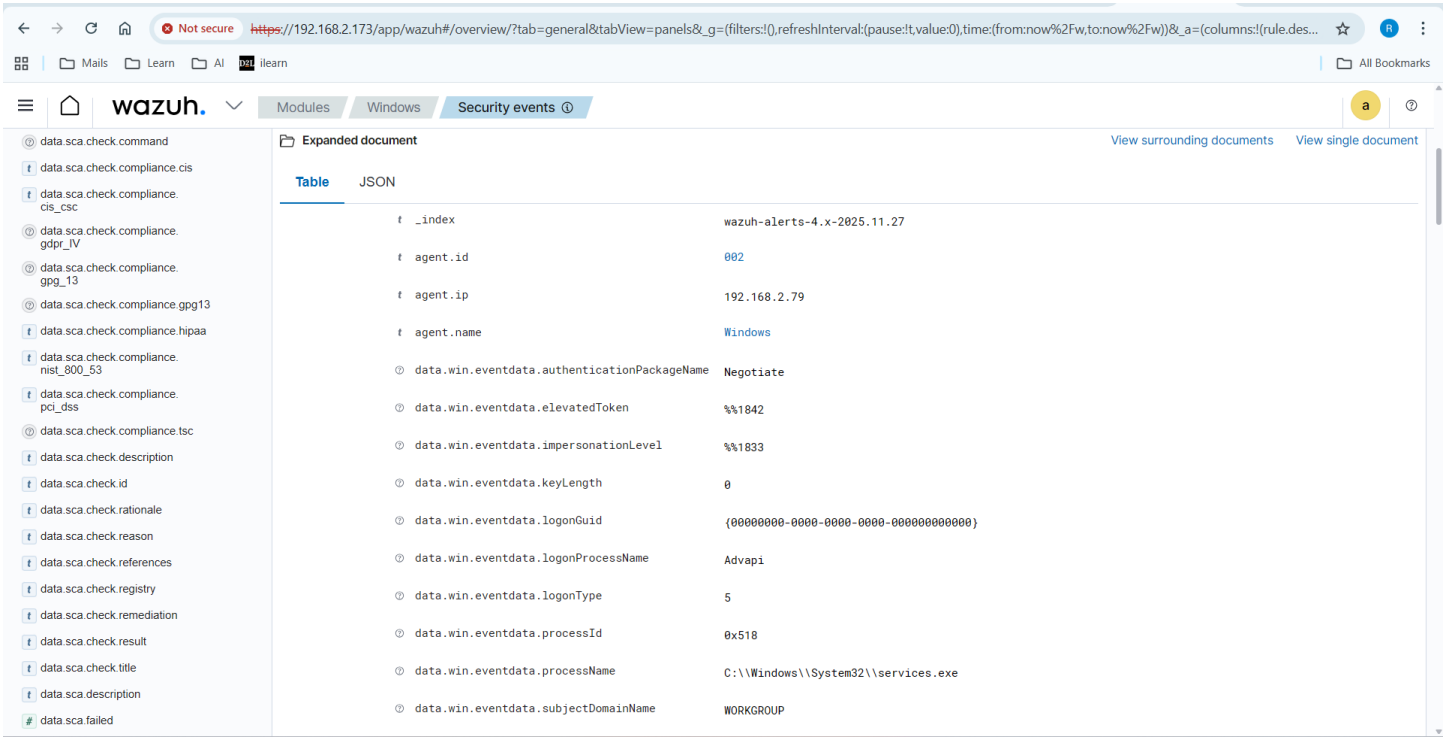


Go to Events option

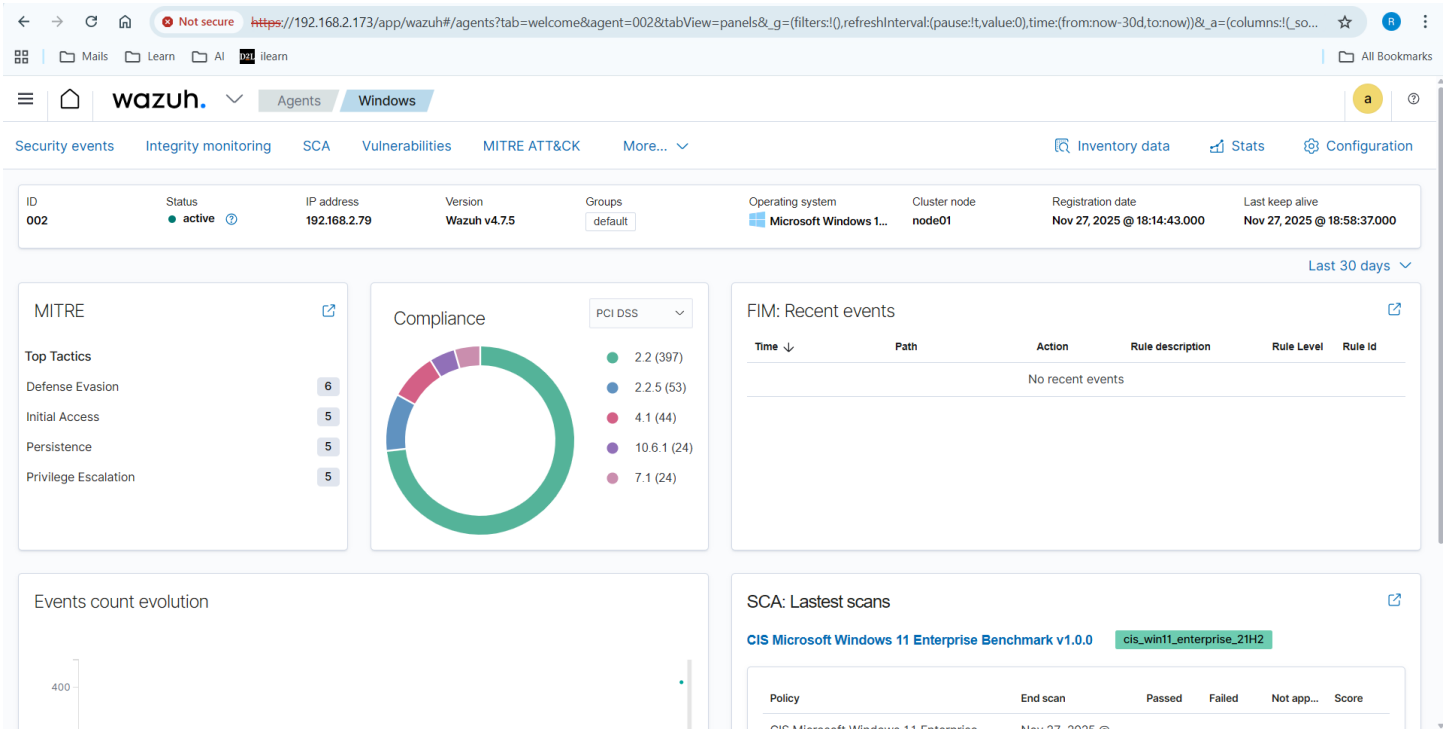
In this page, it provides details on what had happened with the system even removing files are detected as a security event.

We can analyze these events individually by selecting the event's dropdown. You can sort events based upon various factors such as timestamp, tactics, or description.

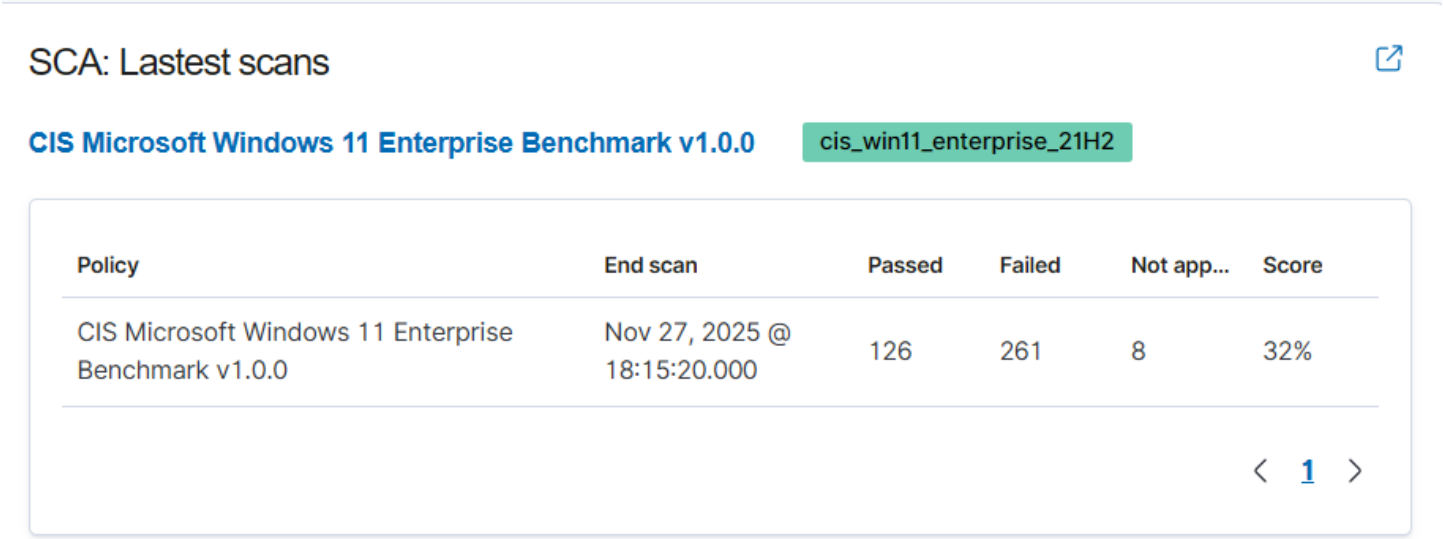





## Policy Auditing:

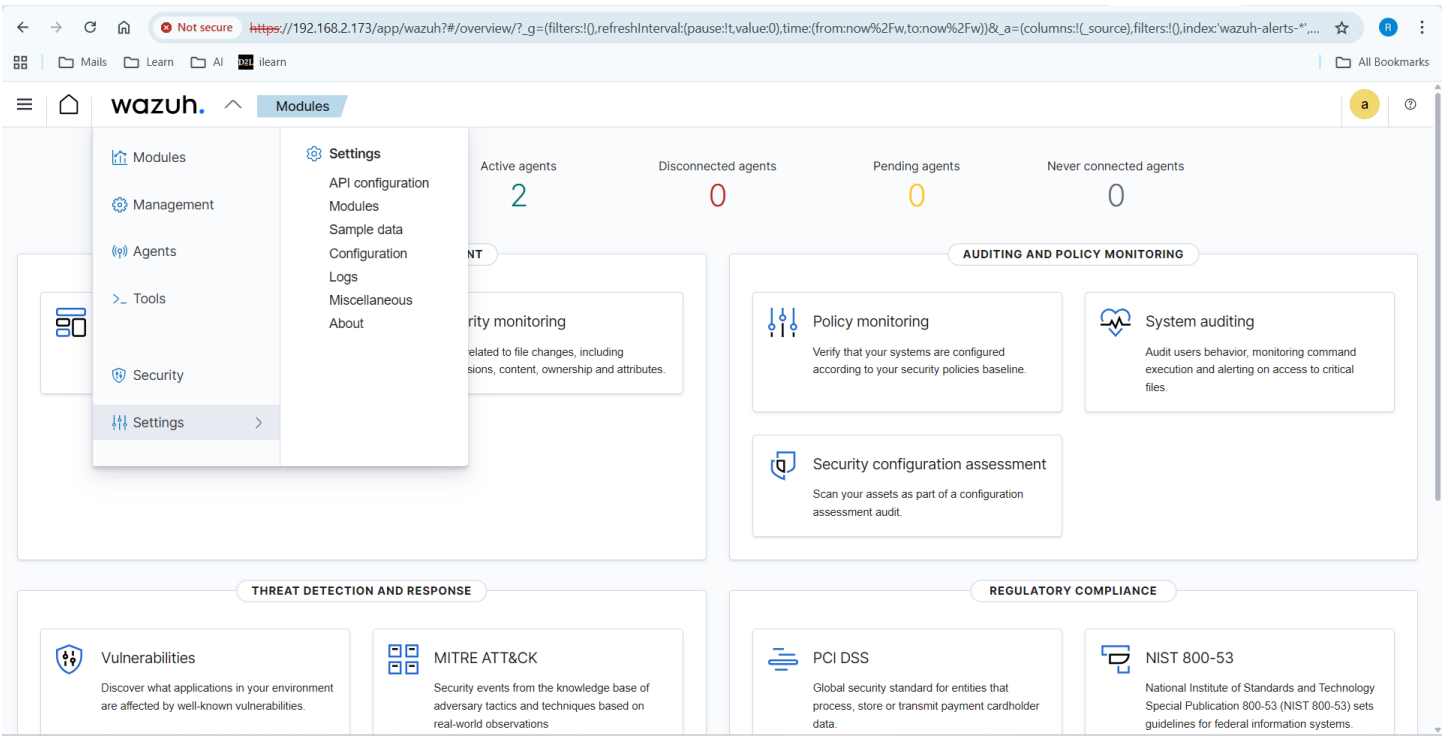


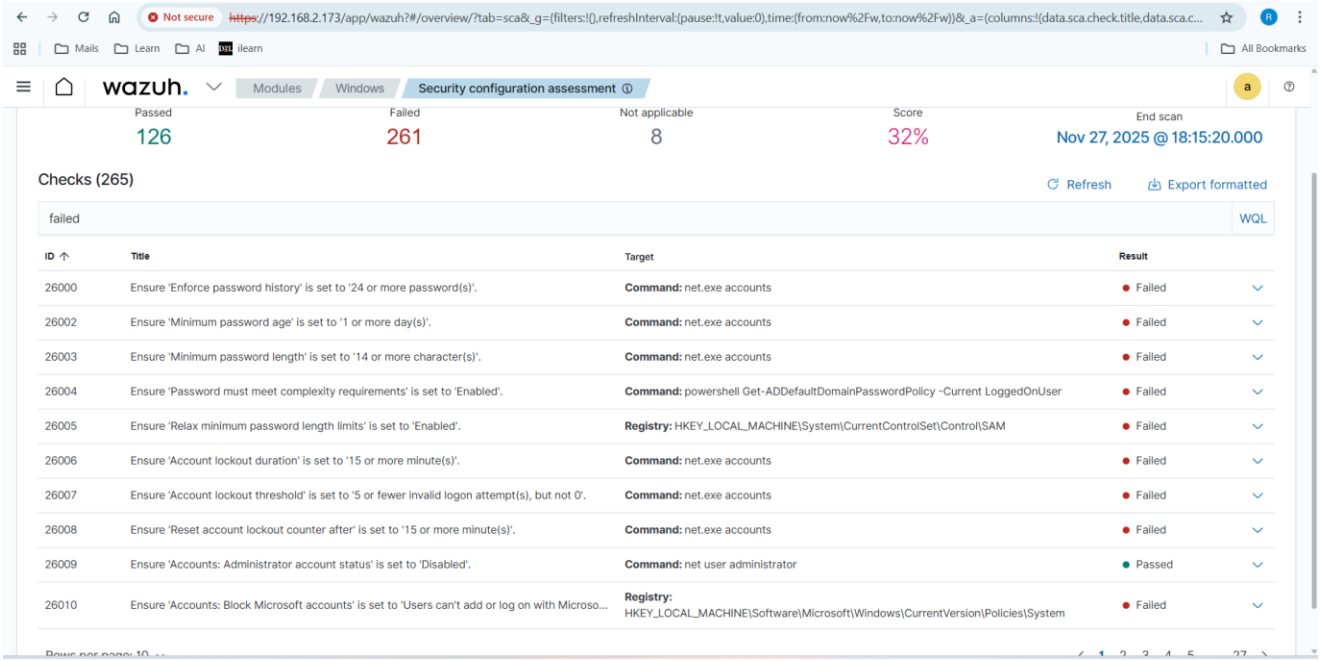
Wazuh presents a broad illustration of the logs. We can use the visualizations to break down this data and explore it further.



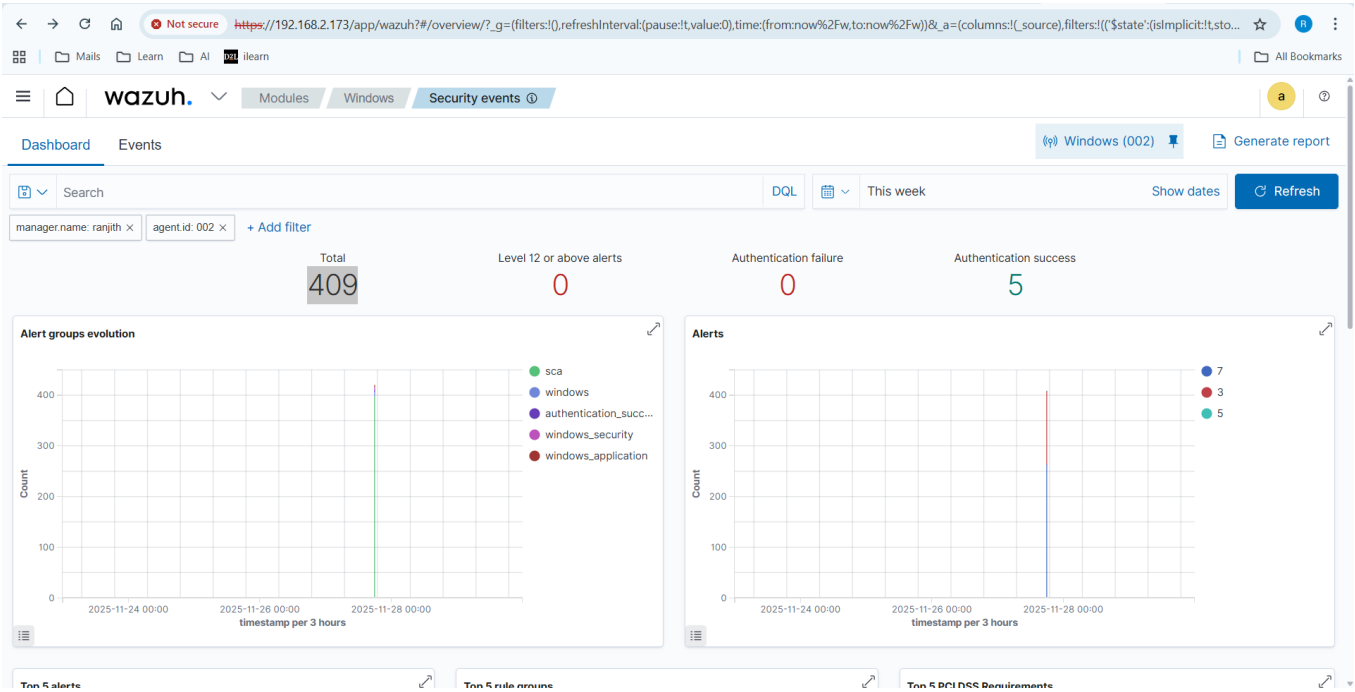

**Monitoring Logons with Wazuh**

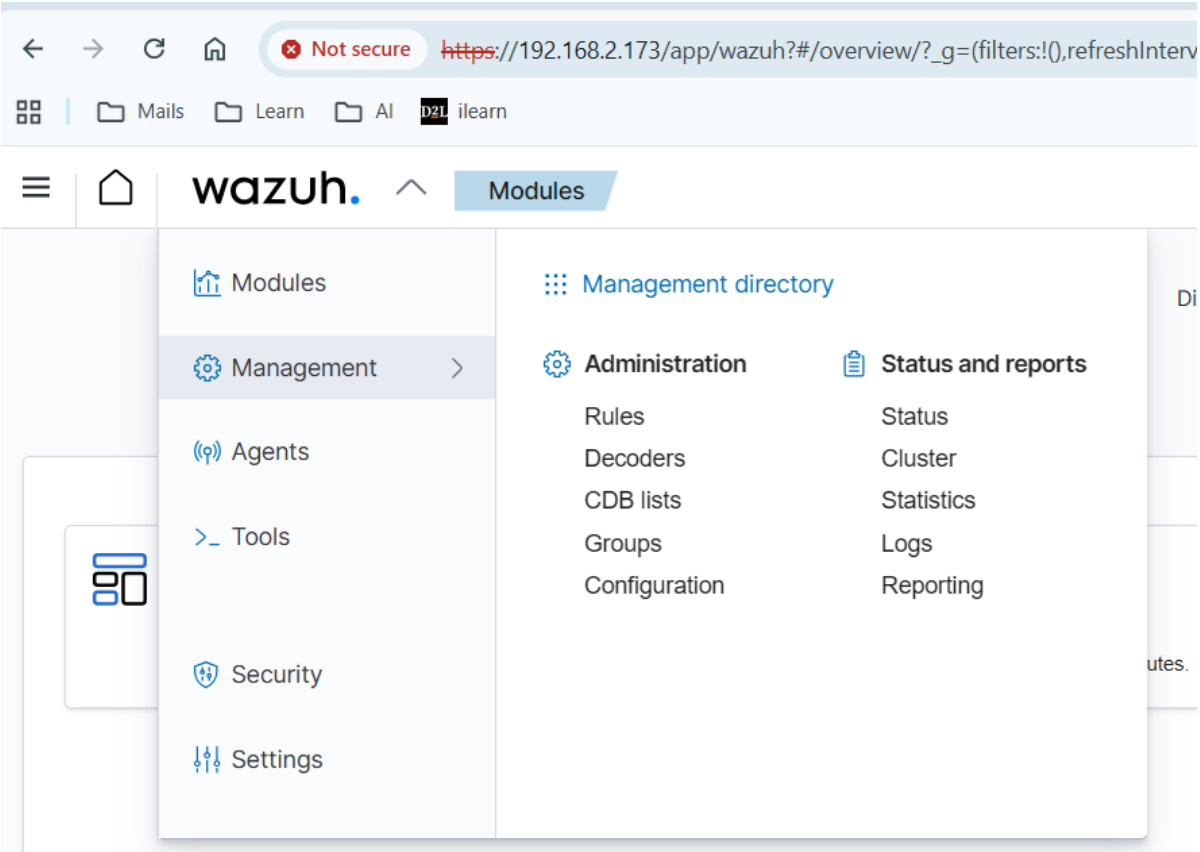
Wazuh's security event monitor is capable of actively recording both successful and unsuccessful authentication attempts.



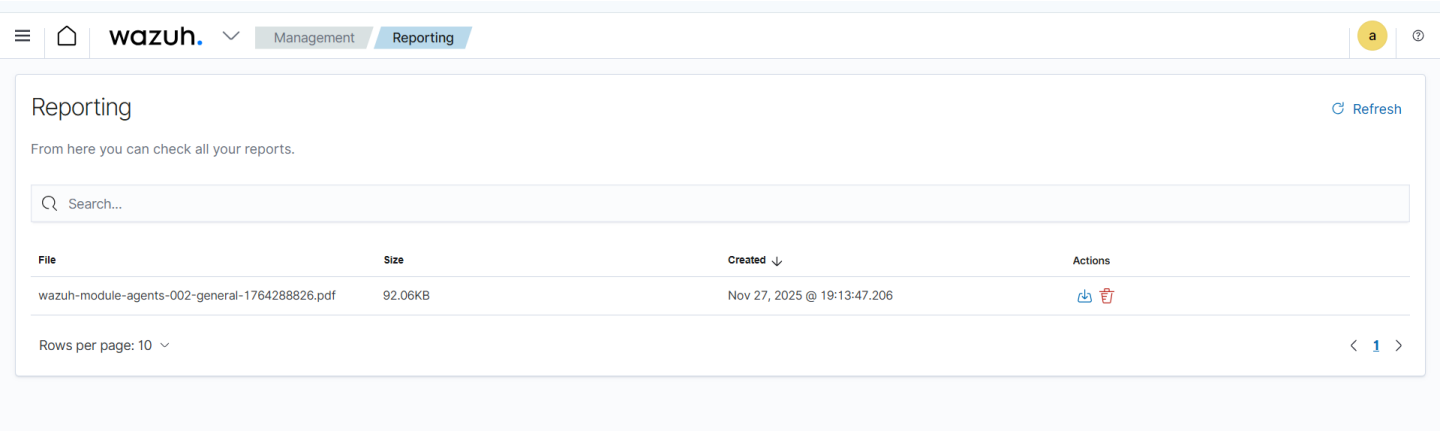



You can also export the filter result to a .csv file which you can use to track the progress of remediations and workaround.






Go to Management and Go to Reporting




- I installed and configured the Wazuh server on the Ubuntu server and configured both agents in Windows and Linux platforms.
- I also looked  Windows Event logs for monitoring. Generating security events reports for clients.
- Wazuh is a powerful open-source security detection tool to identify threats using its alert rule system.